

Student Dropout Prediction: A Machine Learning Analysis

DSAA2011(L01) Course Project - HKUST(GZ) 2026 Spring

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Introduction

This study is based on the “Student Dropout and Academic Success” dataset from the UCI repository. Through a complete machine learning pipeline, we aim to predict students' academic outcomes. This presentation will focus on the following two parts:

1. **Clustering Analysis:** We will present the clustering analysis, detailing the process from *algorithm selection* (addressing the limitations of hard boundaries → choosing GMM), *parameter tuning*, to the *validation of clustering result reasonableness*.
2. **Open-ended Exploration:** We will elaborate on the open exploration conducted to enhance predictive performance.

Clustering (1/3): Algorithm Selection & Tuning

1. Hard-Boundary Models

- ▶ **Exploration:** Tested Hierarchical (Ward) and K-Means.
- ▶ **Optimal K:** $K = 4$ provided the best granularity.
- ▶ **Limitation:** While K-Means performed well, rigid spherical boundaries are less flexible in handling the severe structural overlap of the "Enrolled" class.

⇒ *Conclusion: A probabilistic approach is needed to handle the overlap.*

2. Final Selection: GMM

- ▶ **Best Config:** $K = 4$ with **Tied Covariance**.
- ▶ **Why GMM?** Soft-assignments handled overlap naturally; "Tied" covariance reduced 40D overfitting.

Performance Comparison ($K = 4$)

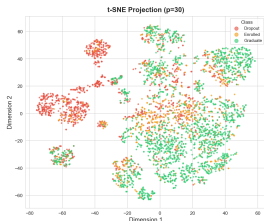
Algorithm	ARI	NMI
Hierarchical (Ward)	0.153	0.149
K-Means	0.313	0.230
GMM (Tied)	0.334	0.273

Clustering (2/3): Core Insight & Visualizations

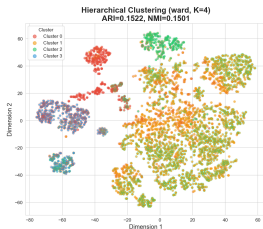
Insight: Natural Label Consistency

- **Consistency:** Clustering results seem to be basically consistent with true labels.
- **Ablation Check:** Removing target-encoded features yielded identical structures.

Original Labels



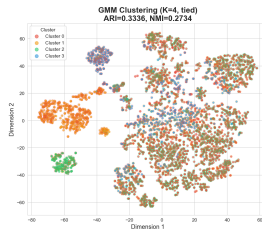
Hierarchical (Ward)



K-Means ($K = 4$)



GMM (Tied)

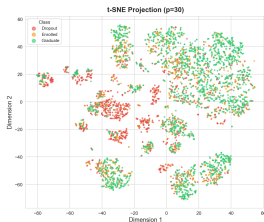


Clustering (3/3): Core Insight & Visualizations

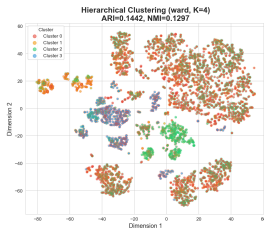
Insight: Natural Label Consistency

- **Consistency:** Clustering results seem to be basically consistent with true labels.
- **Ablation Check:** Removing target-encoded features yielded identical structures.

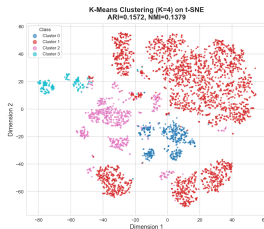
Original Labels



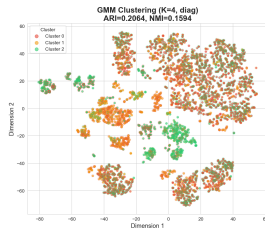
Hierarchical (Ward)



K-Means ($K = 4$)



GMM (Tied)



Open-ended Exploration: LR Class-Bias Tuning

1. Motivation

- ▶ `class_weight='balanced'` overcompensates for Enrolled ($1.86 \times \text{weight}$), causing over-prediction in borderline cases.

2. Method: Boundary Refinement

- ▶ Post-hoc bias: $\hat{y} = \arg \max_k [\log P(y=k | x) + b_k]$
- ▶ Fix $b_{\text{Dropout}}=0$, search only 2 offsets (2-DOF).
- ▶ 41×41 grid on *validation set*.
- ▶ Selected: $b = [0, 0, +0.4]$ — Graduate gets +0.4 log-odds.

3. Significance

- ▶ Bootstrap ($n=10k$): $p = 0.051$ — borderline at 95% level.

4. Model Ranking (Test F1 macro)

Model	Acc	F1
Decision Tree	0.712	0.676
SVM (RBF)	0.732	0.696
Logistic Regression	0.741	0.708
Random Forest	0.747	0.706
ExtraTrees	0.767	0.714
MLP (tuned)	0.776	0.708
LR + Class-Bias	0.765	0.723

5. Per-Class Gain vs LR

	Drop.	Enr.	Grad.
Δ F1	+0.004	+0.010	+0.025

LR + class-bias tuning: #1 among all 9 models.

Conclusion

Key Findings

1. Engineered top-40 features carry strong signal; academic metrics dominate.
2. **Enrolled** is structurally intermediate — not a natural cluster.
3. LR is the strongest baseline; LR + class-bias tuning achieves best F1 = **0.723**.

Limitations & Future Work

- ▶ LR+bias gain is borderline significant ($p = 0.051$)
- ▶ Future: better Enrolled modeling, external validation, early-warning deployment

Reference

Realhino, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>

Q & A

Thank You!